

Can treatment with chondroitin and glucosamine sulphate prevent changes in the articular disc caused by temporomandibular joint osteoarthritis?

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Abstract

Background: Chondroitin and glucosamine sulphates (CGS) are considered structure-modifying drugs and have been studied in the prevention, delay or reversal of structural morphological changes in joints caused by osteoarthritis.

Objective: The aim of the present study was to investigate the action of CGS on the progression of chemically induced osteoarthritis in the temporomandibular joint (TMJ) of rabbits by evaluating the serum levels of tumour necrosis factor (TNF- α) and collagen in the articular discs.

Materials and Methods: A sample of 36 male rabbits was divided into three groups: control (CG), osteoarthritis (OG) and treatment (TG). The disease was induced by intra-articular injection of sodium monoiodoacetate (10mg/mL) in the OG and TG groups bilaterally. After 10days, the TG animals received subcutaneous injection of chondroitin sulphates and glucosamine (7.5mg/kg) and the OG and CG received saline solution (50 μ L). Euthanasia times were subdivided into 40 and 100days. Collagen quantification was performed by biochemical and histological analysis and for the quantification of serum levels of TNF- α , an enzyme immunoassay was used.

Results: The TG showed an increase in the collagen area of the articular disc when compared to the CG and the OG. The increase collagen concentration in the discs did not show a statistically significant difference between the groups. Post-treatment TNF- α levels were significantly lower in TG compared to OG.

Conclusions: The results indicate that CGS treatment delayed the degeneration of the collagen in the TMJ articular disc and reduced serum TNF- α levels, indicating a preventive effect on OA progression.

KEYWORDS

chondroitin sulphate, collagen, glucosamine sulphate, osteoarthritis, temporomandibular joint, TNF- α

1 | INTRODUCTION

Chondroitin and glucosamine sulphates (CGS) are considered slow-acting symptomatic drugs (SYSADOAs) that relieve pain and have good tolerability and safety. These compounds have been tested as structure-modifying drugs (SMD) that may prevent, delay, or reverse morphological structural changes in the joint caused by osteoarthritis (OA).^{1,2}

Temporomandibular joint (TMJ) osteoarthritis leads to cartilage and articular disc degradation, affecting chewing and phonation functions. The temporomandibular joint disc plays a crucial role as a congruent agent, absorber and distributor of stress in the mandibular cinematic complex.³ Part of its mechanical properties have been related to the presence of a highly organised collagen network and a sparse amount of glycosaminoglycans (GAGs) in the solid extracellular matrix (ECM) of articular discs.^{1,4-7} During an inflammatory process, as in OA, an imbalance in cytokines and growth factors present in the synovial fluid is observed, favouring the increase of inflammatory cytokines such as tumour necrosis factor (TNF- α).⁸ This increase induces the synthesis and release of proteases responsible for collagen and proteoglycan depletion, reflecting in the cartilage degradation observed in osteoarthritis.¹

The therapeutic options available for the management of OA are basically restricted to the use of anti-inflammatory drugs and analgesics, but their chronic administration is limited by their deleterious systemic effects.⁹ Moreover, conventional treatments are not satisfactorily effective in controlling OA progression nor in regenerating damaged cartilage. For this reason, new therapeutic alternatives have been investigated for the management of degenerative diseases such as osteoarthritis, seeking not only to control symptoms, but also to preserve joint structures and maintain the patient's quality of life.⁹⁻¹⁴ In the present study, we investigated the therapeutic effect of CGS on the progression of osteoarthritis in the articular disc of the temporomandibular joint of rabbits by analysing serum levels of TNF- α and quantifying disc collagen.

2 | METHOD

2.1 | Experimental design

Thirty-six male New Zealand rabbits (*Oryctolagus cuniculus* L.), adults, 4-months-old, weighing between 3 and 4 kg, were used. The animals were kept under ad libitum feeding based on water, rabbit food and leafy greens, in individual cages, with environments under 12h light/dark cycle, temperature (21°C $\pm$ 1) and relative humidity between 40% and 60%. This study is in accordance with the Law 11794/2008, which establishes the procedures for scientific use of animals, according to current legislation in Brazil. Also, it was submitted to the evaluation and approval of the Ethics Committee on the Use of Animals in Research (CEUA) of the Hospital de Clínicas de Porto Alegre (HCPA) with number 2018-0424.

The rabbits were randomly randomised between the control group (CG, $n=12$), which received only saline solution injection into

the TMJ, the osteoarthritis group (OG, $n=12$) and the treatment group (TG, $n=12$), which received an injection of sodium monoiodoacetate (MIA) into the TMJ, bilaterally. After 10 days, animals in the TG started treatment with subcutaneous injection of chondroitin sulphates and glucosamine. The CG and OG animals received saline solution by the same route. Each group was subdivided and catalogued into three other subgroups, according to the treatment and time of death of the rabbits at 40 and 100 days after the induction of OA (Figure 1). The times of death of the animals were defined based on the studies by Uebelhart et al.¹⁵ and Artuzi, et al.^{1,16}

The animals were anaesthetised with a combination of Ketamine (20mg/kg), Meperidine (3mg/kg) and Midazolam (1mg/kg), intramuscularly. After trichotomy, the zygomatic process was located by digital palpation and the extension of this process to the external ear canal was marked on the skin with methylene blue. Then, 5mm distal and superior to the posterior limit of the zygomatic process, the point of introduction of the needle for the infiltration of solutions was marked. Finally, a solution containing 0.5mg of MIA (Sigma-Aldrich, St. Louis, USA) dissolved in 50 μ L of saline solution (10mg/mL) was slowly injected into the temporomandibular joint, bilaterally, of rabbits in the OG and TG groups. The CG group received the same amount of 50 μ L of saline solution in the TMJ. The osteoarthritis induction model followed the protocol of Artuzi et al.¹ According to the established model, the animals did not present clinical changes indicating suffering, pain or weight loss, and the use of analgesic medication was not necessary at this stage.

2.2 | Treatment

The TG animals received, subcutaneously, an injection of 0.1mL/kg of Chondroton® every 3 days, which corresponds to approximately 7.5mg/kg of chondroitin sulphate and 7.5mg/kg of glucosamine sulphate. The administration of this medication was started 10 days after the induction of OA. The doses were calculated according to the recommended dosage in the drug package insert and according to the dose used in studies previously performed by Artuzi et al.^{1,16} Due to the probable variation in the weight of the animals and the need to adjust the dose of the medication, the animals' weight was controlled on a weekly basis. The times 0, 10, 40, and 100 days corresponded, respectively, to the time at which OA was induced, treatment started, and the animals died. The animals in the CG and OG groups received, subcutaneously, only one volume of saline solution, according to their body weight.

To quantify TNF- α , blood samples (10mL) were collected from all animals, through the lateral auricular vein, at 10, 40 and 100 days. The animals were sedated with Ketamine (20mg/kg), Meperidine (3mg/kg) and Midazolam (1mg/kg), intramuscularly.

2.3 | Death and processing of the specimens

According to the predetermined times (40, 100 days), the animals were anaesthetised with an association of Ketamine (20mg/kg),

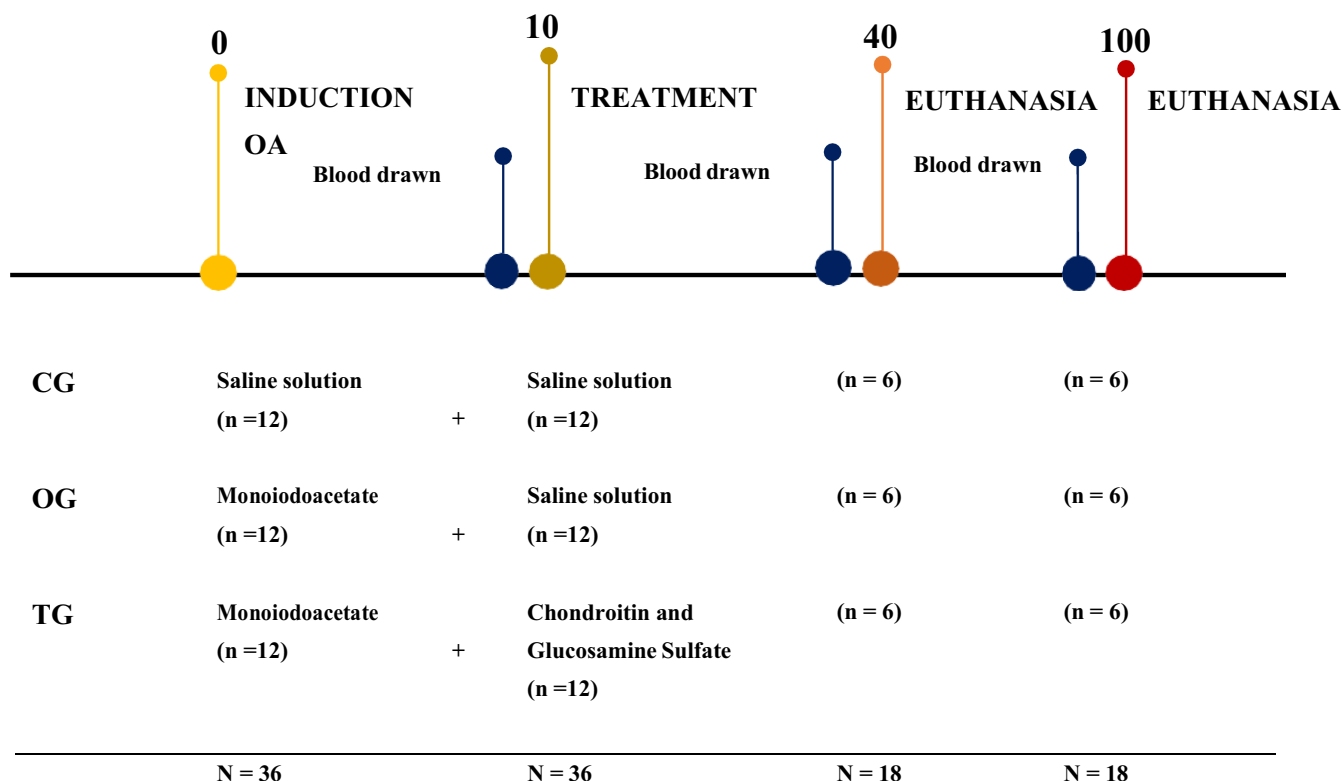


FIGURE 1 Study design. CG–control group, OG–osteoarthritis group, TG–treatment group, OA–osteoarthritis, MIA–sodium monoiodoacetate, CGS–chondroitin and glucosamine sulfate. Thirty-six rabbits were divided into three groups: CG, OG, TG. The disease was induced by injection of MIA in OG and TG groups. After 10 days the TG animals received subcutaneous injection of CGS and the OG and CG groups received saline solution. Blood samples were collected before the animals were euthanized. The sample was subdivided into two euthanasia times, 40 and 100 days.

Meperidine (3 mg/kg), and Midazolam (1 mg/kg), in a final volume of approximately 2 mL, intramuscularly, and killed by anaesthetic overdose of Propofol (5 mg/kg), followed by an intravenous injection of Potassium Chloride (1 mL/kg). The temporomandibular joints were dissected bilaterally. The articular discs (36) on the left side were removed, visualised with a magnifying glass, and preserved at -80°C for collagen quantification. The right-side temporomandibular joints were fixed in 10% formalin and decalcified in 5% nitric acid. Subsequently they were sectioned in the sagittal plane, embedded in paraffin and processed for histological analysis.

2.4 | Quantification of collagen in the articular disc by biochemical analysis

The collagen concentration in the articular discs was determined by the Sircol™ Soluble Collagen Assay (Biocolor, UK), following the manufacturer's instructions. Disc samples were weighed on a precision balance and stored in 1.5 mL Eppendorf Protein Lobin tubes. For extraction of acid-pepsin-soluble collagen, specimens were digested with 0.5 M acetic acid containing 0.1 mg/mL pepsin (P7012; Sigma-Aldrich, USA) at a 1:20 ratio and incubated at 4°C for 48 h. The supernatant was removed after centrifugation at 3000 rpm for 10 min (Hettich–Universal 320R Centrifuge) and transferred to

another Eppendorf tube. To each tube of soluble collagen, 1 mL of Sircol dye reagent was added and mixed by inversion and gentle agitation on a shaker (KJ-201BS Oscillator) for 30 min and centrifuged at 12000 rpm for 10 min. The precipitated dye-collagen complex was washed and drained with salt-acid washing reagent. Soon after, alkaline reagent was added and mixed by vortexing. The absorbance of the samples was measured at 555 nm using a microplate reader (Spectra Max® M3–Hexis Scientific). A collagen calibration standard, provided in the kit, was used to obtain the standard curve (15, 30 and 50 μg). The collagen concentration was measured in $\mu\text{g}/\text{mg}$.

2.5 | Image capture and histological analysis

Sagittal $0.4\mu\text{m}$ sections of the temporomandibular joints along the postero-anterior axis, including the anterior, middle, and posterior portions, were obtained and stained using *PicroSirius Red* according to the manufacturer's protocol (EasyPath). Images were captured with a polarised microscope (Olympus-BX51TF) coupled to a camera device (Olympus, Q-Colour 5). Settings were standardised and captures were taken at 4x magnification, where three fields for each sample were selected and images stored in TIF format.

TABLE 1 Evaluation of the effects of group, time, and group $\times$ time interaction on the variables collagen concentration and total collagen area.

Variables	Control group	Osteoarthritis group	Treatment group	Effects (<i>p</i> -Value)		
	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Group	Time	Group $\times$ Time
Collagen concentration				.380	.001	.826
40 days	15.1 $\pm$ 5.9 ^b	16.6 $\pm$ 3.9 ^b	16.2 $\pm$ 4.9 ^b			
100 days	8.2 $\pm$ 3.3 ^b	12.0 $\pm$ 4.2 ^b	10.6 $\pm$ 4.4 ^b			
Total collagen area				.014	.708	.151
40 days	0.45 $\pm$ 0.15 ^a	0.36 $\pm$ 0.21 ^a	0.63 $\pm$ 0.09 ^a			
100 days	0.36 $\pm$ 0.10 ^a	0.49 $\pm$ 0.15 ^a	0.55 $\pm$ 0.06 ^a			

Note: Bold values are significant.

^aBetween-group analysis.

^bIntragroup analysis. Equal letters do not differ by Fisher's Least Significant Difference (LSD) test at 5% significance between groups.

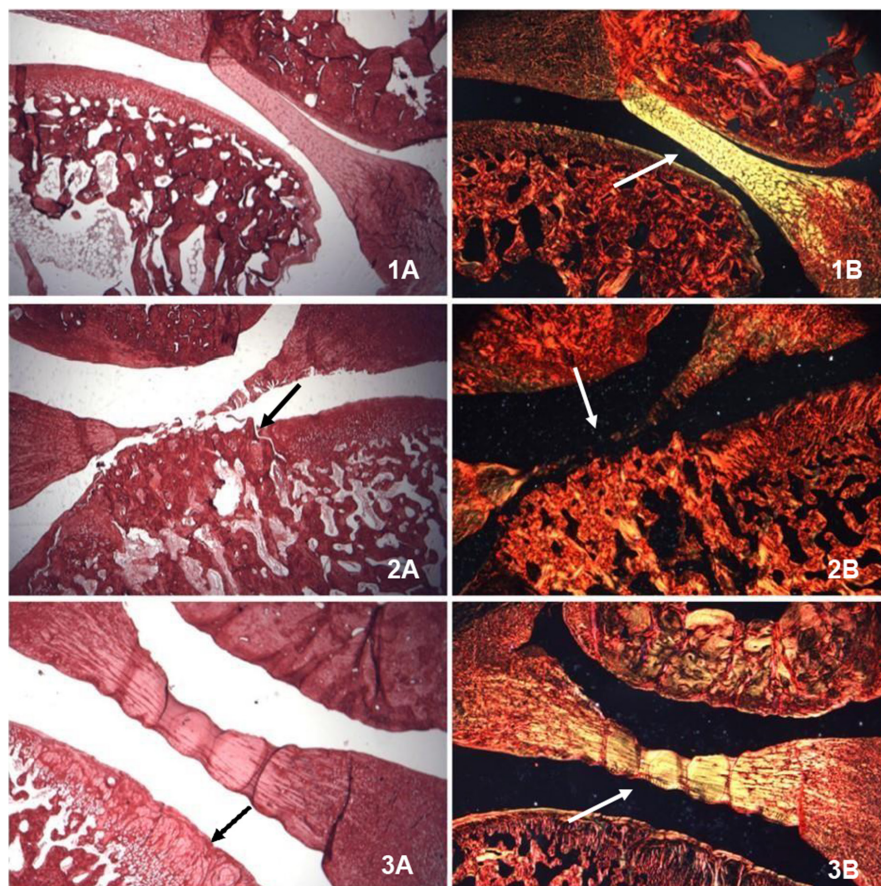


FIGURE 2 Histological analysis of the collagen area occupied in the articular disc of rabbits 40 days after treatment. Images in 4 $\times$ objective. Picrosirius Red staining of rabbit TMJ sagittal sections with the structures indicated. Images were obtained under non-polarised light (A) and polarised light (B). (1) Control group (CG), (2) osteoarthritis group (OG), (3) treatment group (TG). The better collagen organisation correlates with the CGs TG post OA induction and the CG. The white arrowheads indicate the articular disc. In image 1B and 3B the disc is intact and the collagen fibres are preserved. The white arrow on image 2B indicates the rupture of the collagen fibres. The black arrow in image 2A shows the presence of subchondral fissure, erosion and discontinuity of the articular surface. The black arrow in image 3A (TG) shows changes caused by OA, such as reduced cartilage layer thickness, fibril formation, and cell degeneration. However, the articular disc, white arrow image 3B, remains intact.

The total collagen area (cm²) of the articular disc was measured on polarised images using ImageJ 1.53k software (Wayne Rasband and contributors, NIH, USA) using a semi-automatic technique. First, the measurements were calibrated from pixel to cm in the software and the size of the image selected for analysis was manually standardised to an area measuring 3 cm long $\times$ 0.06 cm high for all samples. Subsequently, the entire occupied area per disk was selected (by the free hand option) and the 8 bit type and threshold setting was selected to invert the background into black and white. Finally, particle analysis was performed (0-infinity, overlay) and the data was tabulated and analysed in cm² of collagen.

2.6 | Quantification of TNF- α by serum analysis

An enzyme immunoassay, Rabbit TNF- α ELISA (MyBioSource) was used for the quantification of the concentrations of serum TNF- α levels of the samples. The amount of 100 μ L of the standard and sample was added to each well of the plate pre-coated with specific antibody provided by the kit and incubated for 90 min at 37°C. Immediately after, 100 μ L of Biotinylated Detection was added and incubated for 1 h at 37°C. The wells were washed and aspirated five times in an automated way (Asys), 90 μ L Substrate Reagent was added and incubated for 15 min at 37°C followed by the addition of 50 μ L Stop Solution. The

colour change was measured spectrophotometrically at a wavelength of 450 nm (Spectra Max® M3–Hexis Científica), and the values were determined in pg/mL and compared to the standard curve.

2.7 | Statistical analysis

Variables were described by mean and standard deviation and compared over time and between groups using two-way Analysis of Variance (ANOVA) for analysis of collagen concentration and area of collagen occupied in the disc. An Analysis of Variance (ANOVA) for repeated measures, followed by *Fisher's Least Significant Difference* (LSD) test to locate significant differences for analysis of TNF- α levels. In addition, the partial eta square (η^2_p) was calculated to measure the effect size. According to Cohen, values ≥ 0.3 indicate small effect size, between 0.31 and 0.5 moderate effect size, and above 0.5 large effect size. The significance level adopted was 5% ($p < .05$) and analyses were performed using SPSS (SPSS Inc., Chicago, IL, USA) version 21.0 software.

3 | RESULTS

3.1 | Area occupied by collagen in the articular disc

There was a statistically significant effect of group (treatment) ($F_{(2,29)} = 5.163$, $p = .014$), where it was observed that, regardless of the time assessed, there is an increase in the total area of collagen

(cm^2) in the TG when compared to the others, which did not differ (Table 1). We also observed a treatment effect, where there was an increase in the total collagen area (cm^2) in the articular disc of the TG, when compared to the others. Furthermore, TG showed histologically greater preservation of joint structures and better organisation of collagen fibres when compared to OG, which showed histological signs of disc degeneration and less collagen (Figures 2 and 3). There was no time effect in the groups studied ($F_{(1,24)} = 0.144$, $p = .708$) or interactions between the Group and Time ($F_{(2,24)} = 2.051$, $p = .151$).

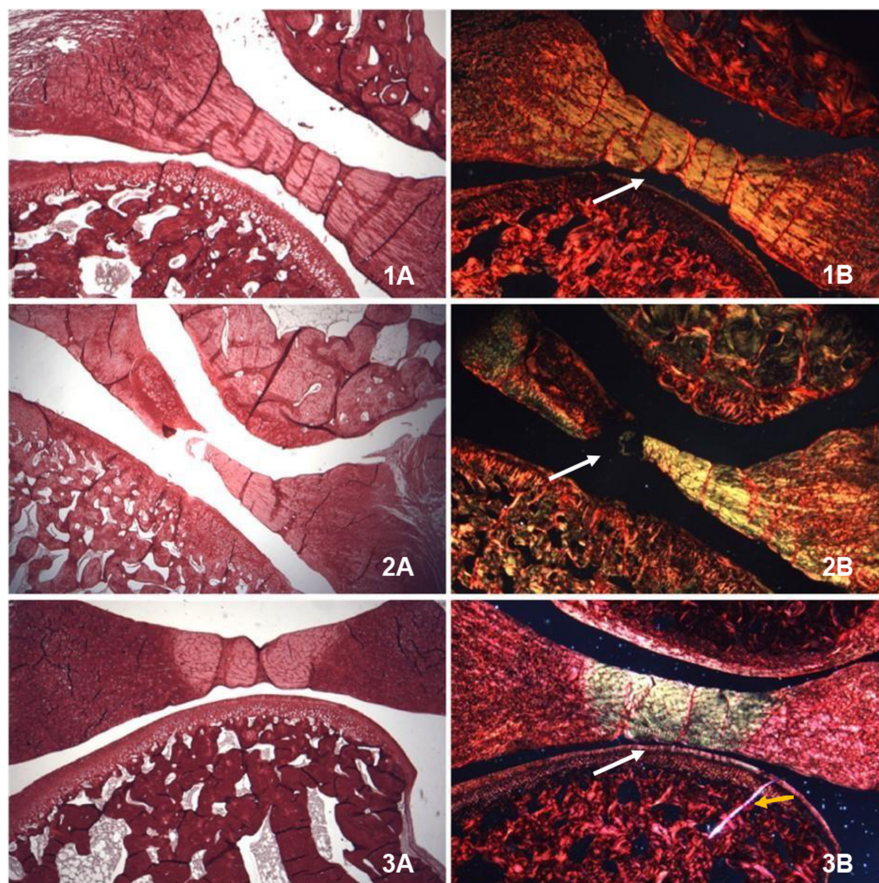
3.2 | Collagen concentration ($\mu\text{g}/\text{mg}$)

There was no significant group $\times$ time interaction effect ($F_{(2,34)} = 0.192$, $p = .826$), as shown in Table 1, that is, the behaviour of the groups was similar between times. When only the effect of time was evaluated ($F_{(2,34)} = 13.796$, $p = .001$), a statistically significant reduction was observed between the times 40 (15.9 ± 4.7) and 100 days (10.3 ± 4.0) ($p = .001$) with the passage of time. Regarding the groups, no significant differences were seen between the treatments ($F_{(2,34)} = 1.001$, $p = .380$).

3.3 | Serum levels of TNF- α (pg/mL)

The Table 2 presents the results of ANOVA for repeated measures in the evaluation of TNF- α levels between groups and time. There was a

FIGURE 3 Histological analysis of the collagen area occupied in the articular disc of rabbits 100 days after treatment. Images in 4x objective. Picrosirius Red staining of rabbit TMJ sagittal sections with the structures indicated. Images were obtained under non-polarised light (A) and polarised light (B). (1) Control group (CG), (2) osteoarthritis group (OG), (3) treatment group (TG). The collagen fibres, which are yellowish green or red, can be seen in quantity and distribution that are better in the CG and TG than in the OG, indicating the preservation of structures after the induction of disease, as indicated in the statistical analyses presented. The white arrows in images 1B (CG) and 3B (TG) show the intact articular disc and preserved collagen fibres. The yellow arrow in image 3B indicates for an imaging artefact. The white arrow on image 2B (OG) shows disruption of the articular disc.



significant interaction effect between group and time, showing that the behaviour between groups was significantly different over time ($F_{(2,30)} = 6.506$, $p = .004$) $\eta^2 = .303$. This is mainly due to the significant increase in TNF- α levels in the OG group when compared to the CG ($p = .002$) and TG ($p = .041$), (Figure 4). In addition, the OG group showed similar results to the CG, with no significant differences ($p = .257$), indicating a reversal of the increase in TNF- α in the OG group after the intervention. We observed an effect of time between pre and post treatment ($F_{(1,30)} = 7.946$, $p = .008$) $\eta^2 = .209$.

4 | DISCUSSION

In the present study, collagen levels were analysed in the TMJ articular disc of rabbits that had chemically induced osteoarthritis and were subsequently treated with CGS. The increase in collagen area observed in TG compared to OG and CG suggests a preventive effect of CGS on osteoarthritis progression, since after inflammation induction it maintained the quantity and organisation of collagen fibres in the articular discs, indicating delayed degeneration of the structures. This is the first study to evaluate the structural effect of CGS in articulation, observing the collagen composition of the disc, a fundamental component for TMJ functionality.

Glucosamine and chondroitin are structural molecules of the disc and articular cartilage and are fundamental for the synthesis

of proteoglycans and glycosaminoglycans (GAGs).¹⁷⁻¹⁹ Studies show that changes in structure and composition in the disc alter its nanomechanical properties and affect its ability to absorb loads during daily activities such as chewing, talking, and yawning.^{3,5,7,20} In addition, the arrangement of collagen fibres in the articular disc influences force distribution, helping to reduce stress concentration in the directly loaded region.^{5,6} In 2019, Cui et al, showed that the presence of inflammation causes a disarrangement and loosening of the collagen molecule, generating a porosity between its bundles, reducing the modulus of elasticity of the inflamed disc region and impairing the ability to resist external stress.²⁰ For this reason, the maintenance of the organisation of collagen fibres of the articular disc has been shown to be as important as its biochemical composition.⁶ The bioavailability of a substrate for the synthesis of proteoglycans and GAGs through CGS supplementation, while joint tissues are exposed to the inflammatory processes of osteoarthritis, would be an interesting alternative, because besides its symptomatic benefits, it would promote a protective effect of the structures. Moreover, OA treatment with CGS in TMJ has shown better effects in reversing degenerative changes than in other joints, possibly due to the presence of mesenchymal cells in fibrocartilage, evidencing its potential as DME.^{1,21} In 2020, Bi et al, observed the existence of a specific population of stem cells in human condyle fibrocartilage, whose chondrogenic potential was regulated by SOX9 expression. In patients with TMJ osteoarthritis

TABLE 2 TNF- α intra and inter-group analyses between pre-treatment versus post-treatment times.

Variables	Control group	Osteoarthritis group	Treatment group	Effects (p -Value; η^2 partial)		
	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Group	Time	Group $\times$ Time
TNF- α				($p = .130$; $\eta^2_p = .127$)	($p = .008$; $\eta^2_p = .209$)	($p = .004$; $\eta^2_p = .303$)
Pre	0.08 $\pm$ 0.13 ^a	0.02 $\pm$ 0.03 ^a	0.09 $\pm$ 0.15 ^a			
Post	0.05 $\pm$ 0.07 ^a	0.32 $\pm$ 0.28 ^a	0.14 $\pm$ 0.17 ^a			

^aEqual letters do not differ by Fisher's Least Significant Difference (LSD) test at 5% significance level. There was a statistically significant increase in TNF- α levels in the Osteoarthritis group over time, while the Treatment group showed similar results to the Control group, demonstrating a prevention in the increase of TNF- α .

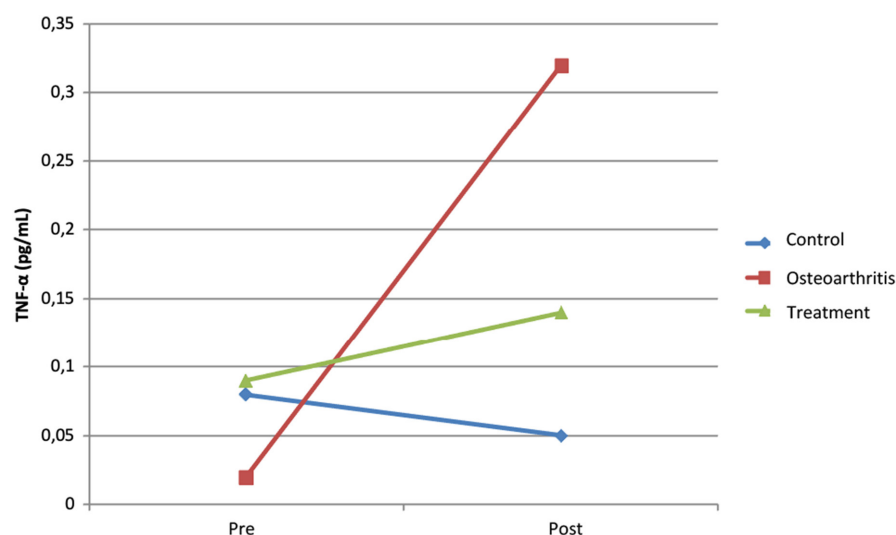


FIGURE 4 Behaviour of serum TNF- α levels between groups during the pre- and post-treatment times. An interaction between group and time is observed. There was a significant increase in the serum levels of TNF- α in the OG*, which did not occur in the other groups, demonstrating a preventive effect of CGS on the increase in TNF- α levels.

who had damaged fibrocartilage, the expression of SOX9 had been suppressed. Interestingly, inflammatory cytokines such as TNF- α activate nuclear factor (NF)- κ B, whose signalling pathway may regulate the expression of this gene.² In the present study, CGS were shown to reduce TNF- α levels even after induction of inflammation, which could indicate their influence in preventing tissue degradation by avoiding SOX9 suppression.

To evaluate the collagen concentration between groups, a biochemical analysis was performed and statistically significant differences were found between times 40 and 100, where the former showed higher collagen concentrations. This can be explained by the increased synthesis of ECM components, such as proteoglycans, in the early stages of OA as an attempt at tissue repair.²² In relation to the groups, although the collagen concentration (μ g/mg) of the TG₁₀₀ and TG₄₀ showed higher concentrations when compared to OG₄₀ and CG₁₀₀, the difference was not significant, possibly due to the small sample size between the subgroups.

In the present study, TNF- α levels increased significantly in the OG group, which had the disease induced and was treated only with saline solution. The same group also showed histological signs of disc degeneration and less collagen. This result is in agreement with previous studies showing increased levels of TNF- α in the synovial fluid of patients with temporomandibular disorders.¹⁷ Moreover, serum levels of TNF- α after treatment with CGS reached levels close to those found in the CG group and both were lower than in the OG group, showing that the treatment was effective in reversing the increase in TNF- α in the TG to levels close to the pre-intervention time, suggesting an anti-inflammatory effect of CGS. TNF- α is considered a protagonist in degenerative processes, because it acts as an initiating cytokine and activates signalling cascades, recruiting immune cells and activating neutrophil migration and the expression of proteolytic enzymes.^{23,24} The positive effects of CGS on TNF- α levels have been reported to modulate inflammation of joint tissues affected by osteoarthritis and stimulate tissue recovery by proteoglycan synthesis and inhibition of NF- κ B.^{1,9,11,15,17,25,26} Activation and translocation of NF- κ B is associated with increased expression of proteolytic enzymes and pro-inflammatory cytokines by chondrocytes and synoviocytes, and the anti-inflammatory effects of CGS are reported to interfere with this process.^{27,28}

It is important to point out that the clinical efficacy of CGS is still controversial, since the studies are highly heterogeneous, with different study designs, dosages, and product presentation, making it difficult to reach a precise conclusion about their effects.²⁹ The routes of administration, oral and intravenous, have different bioavailability and the formulations differ between countries, where in the USA it is considered a nutritional supplement and in some European countries, a medication subject to quality controls.^{12,25,30,31} On the other hand, the advantages of CGS over other medications are: rare gastrointestinal and cardiac adverse reactions, as observed in anti-inflammatory drugs; preservation and recovery of joint tissues, improving jaw function, and consequently, postponing the need for surgical interventions.

It should also be noted that studies in patients are analysed at various stages of disease, tissue repair and different ages, with great variability in

samples. The animal model, as in the present study, allows standardisation of sample characteristics, such as weight, age and sex, as well as the times of disease induction and treatment. Rabbit TMJ is anatomically and histologically similar to the human joint. The rabbit condyle is covered by secondary cartilage and fibrous tissue. Furthermore, similar distribution of collagen and proteoglycans is observed. The most notable morphological difference between rabbit and human TMJ is the shape of the articular surface of the condyle and the retrodiscal area, since the animals do not have a posterior glenoid wall, which facilitates access and manipulation for experiments without the need for surgical access. For this reason, the rabbit is a good animal model for experimental studies of human TMJ diseases and therapies.^{16,32}

In conclusion, the present study showed that CGS can influence the treatment of OA in rabbit TMJ, reducing the serum levels of TNF- α and increasing the amount of collagen in the articular discs, indicating a preventive and anti-inflammatory effect. However, further studies that are TMJ-focused and have similar experimental protocols are needed to corroborate the effects of CGS.

AUTHOR CONTRIBUTIONS

F.A., D.P., E.P. conception of the study and design of the experiments. G.M., F.A., conducting the experiments. A.Q., data analysis. G.M., D.P., A.Q., writing the manuscript. All authors reviewed the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict of interest.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/joor.13814>.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available upon request from the corresponding author.

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